Remote Photoplethysmography

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M1 VIBOT
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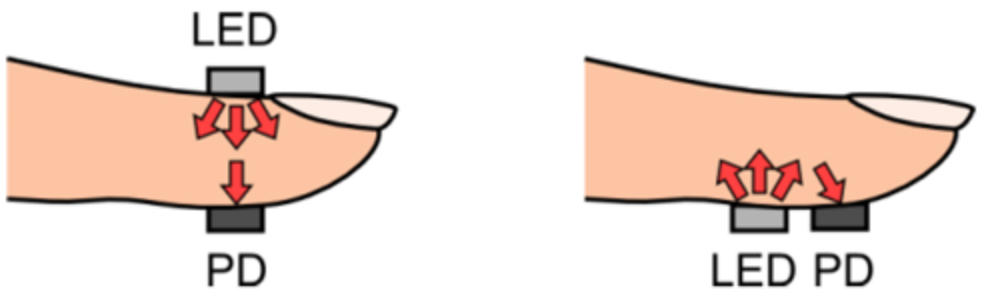
Background on photoplethysmography

Photoplethysmography (PPG)

- a low-cost non-invasive optical technique
- detect microvascular blood volume changes in tissues beneath the skin which are due to the pulsatile nature of the circulatory system.

PPG requires a light source and a photo detector. The light source illuminates the tissue, and the photo detector senses the small variations in reflected or transmitted light intensity.



PPG in transmission mode

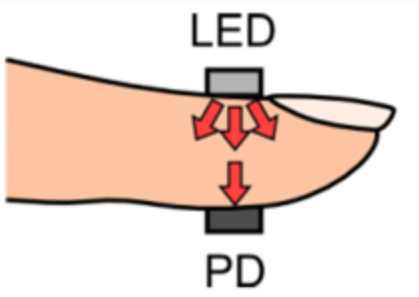
PPG in reflection mode

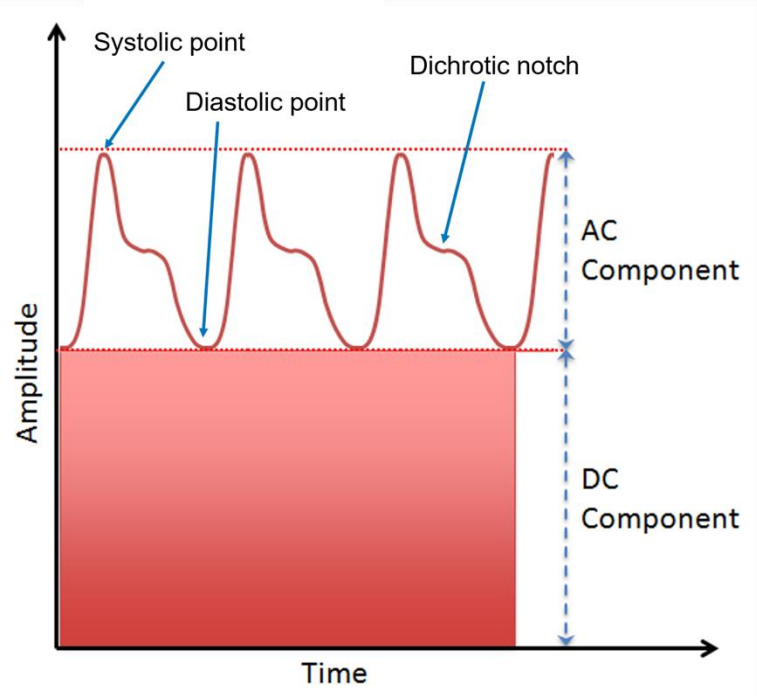
Background on photoplethysmography

Attenuation of light by the finger can be split into:

- Pulsatile arterial blood (AC)
- Non pulsatile arterial blood + venous blood + other tissues (DC)

The AC component is the photoplethysmogram





Background on photoplethysmography

Usually PPG is acquired with pulse oximeters

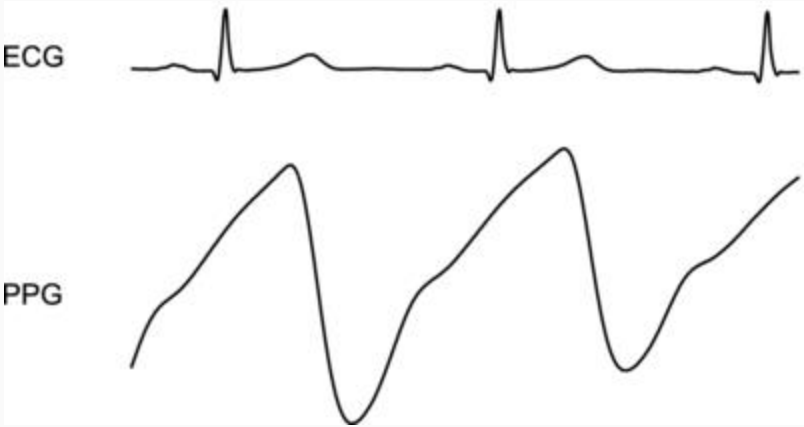
Oxygen saturation

Heart Rate

Photoplethysmogram

Background on photoplethysmography

- Heart rate can easily be estimated from photoplethysmogram



For example with:

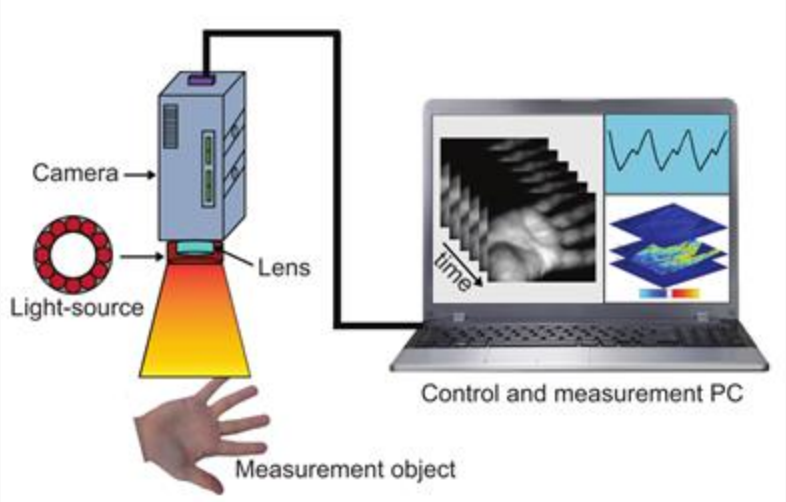
$$Instant\ Heart\ Rate\ (bpm) = \frac{60000\ (ms)}{InterBeatInterval\ (ms)}$$

Remote photoplethysmography

Remote PPG (rPPG) works on the same principle as PPG

Various possible setups

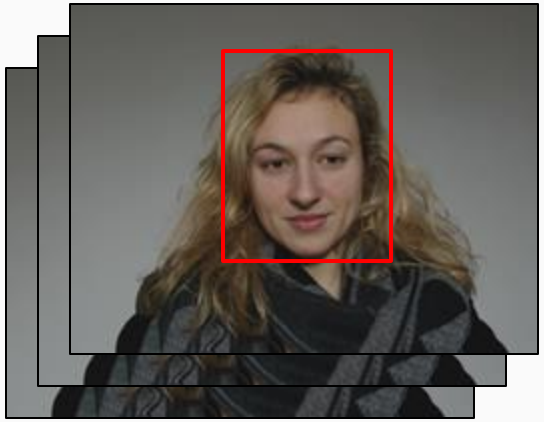
- The photo-detector can be a professional camera or a simple webcam
- The light source can be just ambient light or dedicated light source



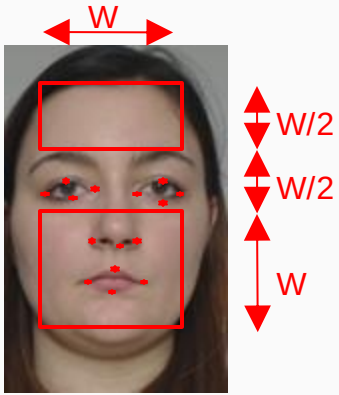
Remote photoplethysmography

Video and signal processings

Spatial ROI selection and tracking → Spatial averaging → Color channel selection - combination → Denoising → Physiological feature extraction



Face detection and tracking



Facial landmark detection for ROI selection

Skin detection

Remote photoplethysmography

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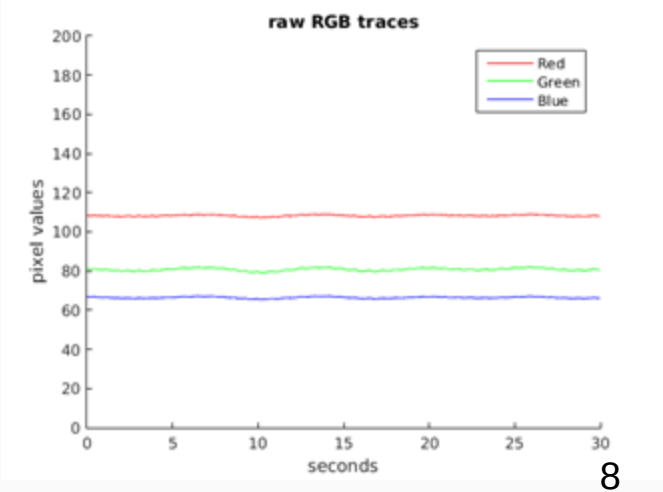
$$[\overline{R}, \overline{G}, \overline{B}]_i$$



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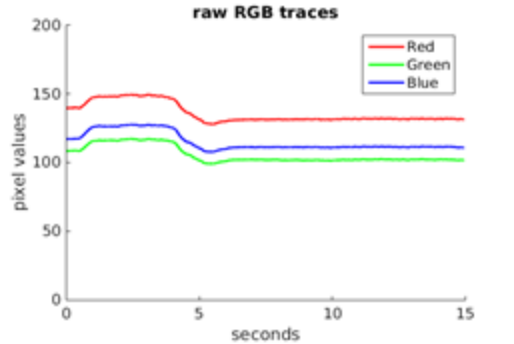
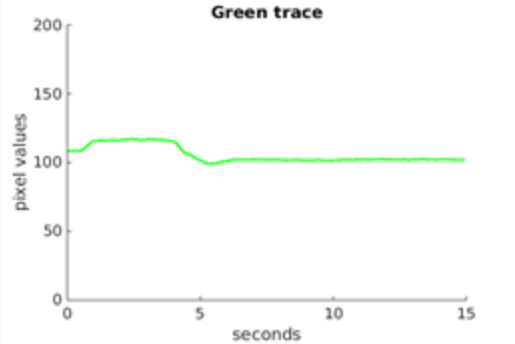
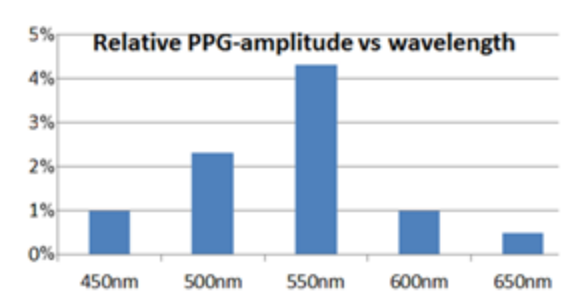
Method 1: Green channel selection





The amplitude of the PPG-signal in light reflected from the skin varies as a function of λ.
Peak around 550 nm (Green channel).



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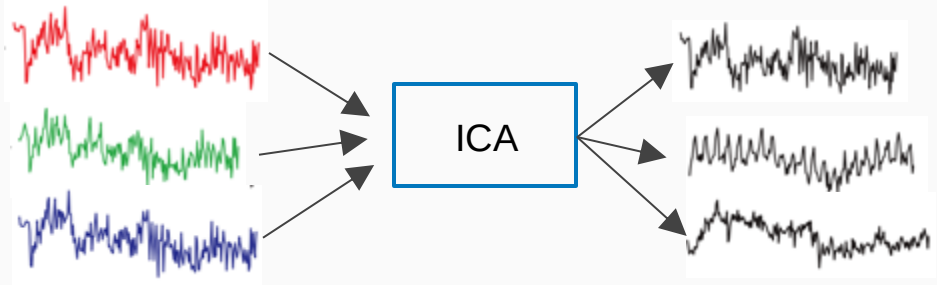
Method 2: Linear combination of RGB channels $$S = \alpha_1 R + \alpha_2 G + \alpha_3 B$$

Independent Component Analysis (ICA): RGB traces are a mixture of PPG along with other sources (motion, changes in ambient lighting conditions).



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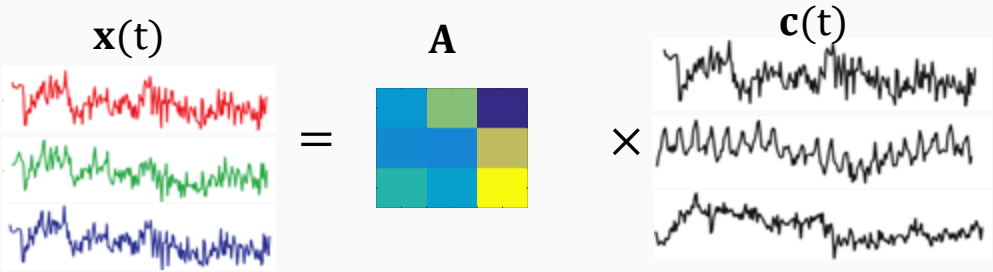
Independent Component Analysis



$$\mathbf{x}(t) = \mathbf{A} \cdot \mathbf{c}(t)$$

$$\mathbf{y}(t) = \mathbf{W} \cdot \mathbf{x}(t)$$

ICA decomposes a multivariate signal into independent signals by optimizing an objective function

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Video and signal processings

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Method 3: Linear combination of RGB channels $$S = \alpha_1 R + \alpha_2 G + \alpha_3 B$$

Chrominance-based method: create 2 orthogonal chrominance signals (X and Y) from color channels via linear combination (~similar to U and V).

The variations due to the blood volume changes in the skin will likely be different in each channel while the motion affects both chrominance signals identically -> we use ratio of the 2:

$$S = \frac{X}{Y} \quad \Rightarrow \quad S = 3\left(1 - \frac{\alpha}{2}\right) R - 2\left(1 + \frac{\alpha}{2}\right) G + \frac{3\alpha}{2} B \quad \text{with} \quad \alpha = \frac{\sigma(X)}{\sigma(Y)}$$

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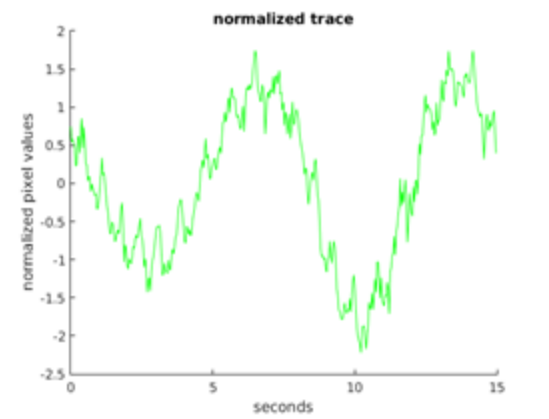

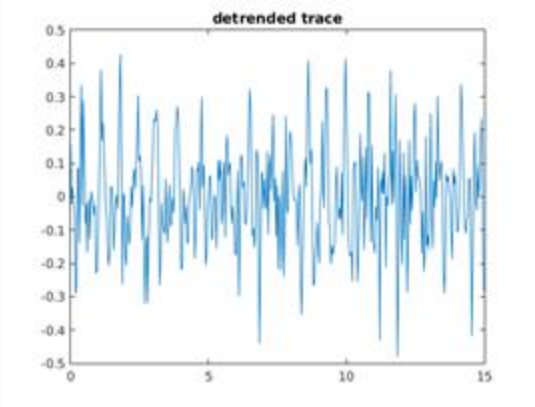

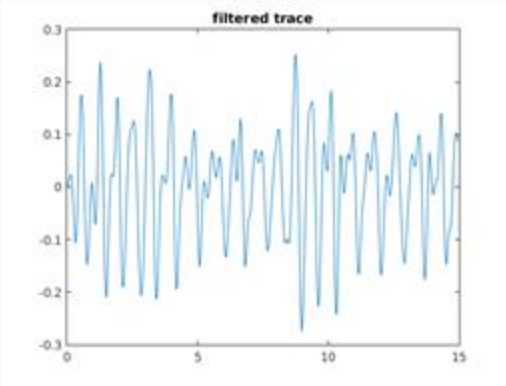

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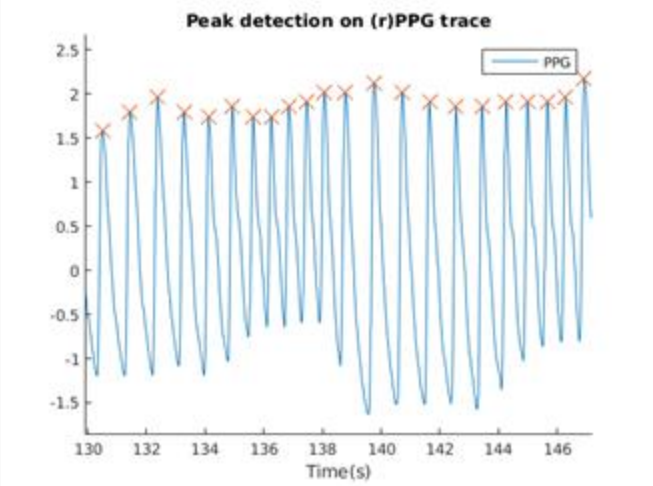
Video and signal processings

Spatial ROI selection and tracking → Spatial averaging → Color channel selection - combination → Denoising → **Physiological feature extraction**

Pulse counting via temporal domain heuristics



$$Instant\ Heart\ Rate\ (bpm) = \frac{60000\ (ms)}{InterBeatInterval\ (ms)}$$

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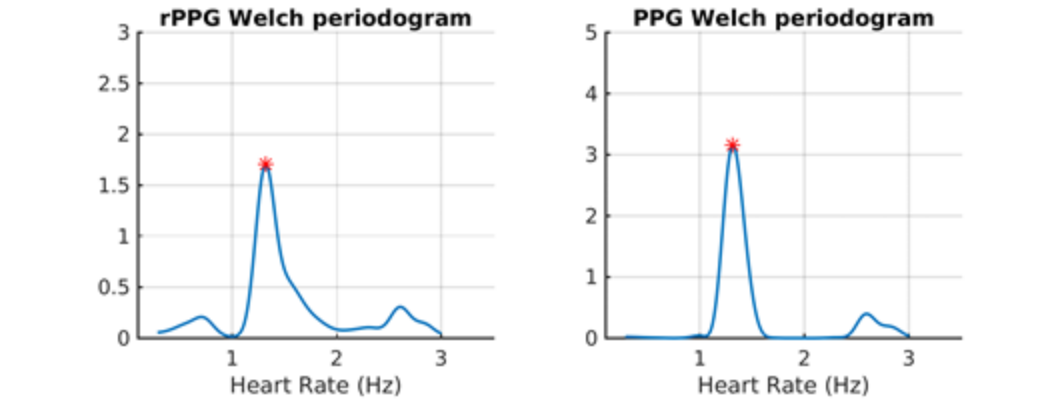
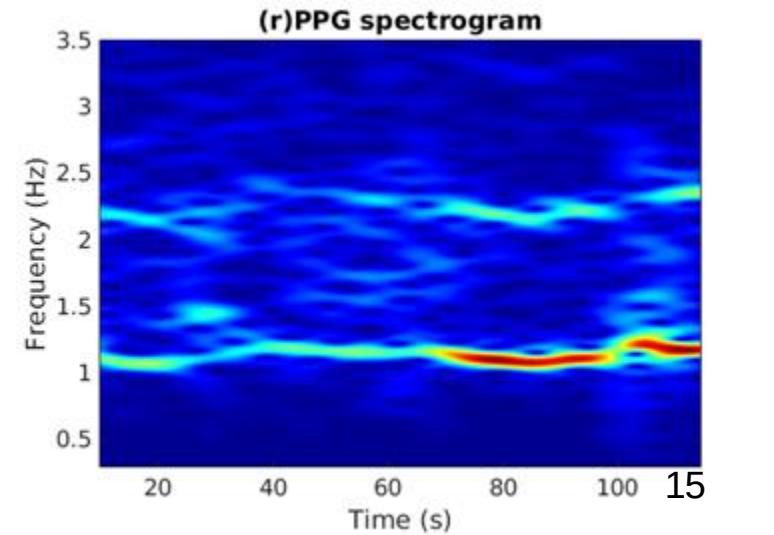
Video and signal processings

Spatial ROI selection and tracking → Spatial averaging → Color channel selection - combination → Denoising → **Physiological feature extraction**

Frequency domain HR estimation





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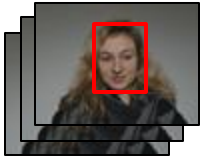
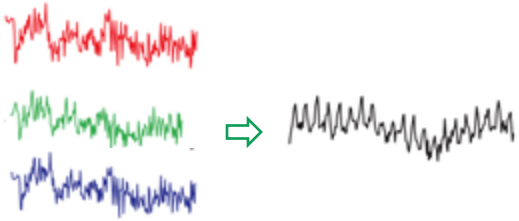
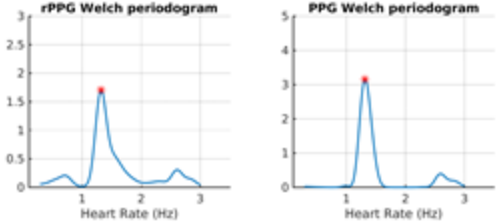


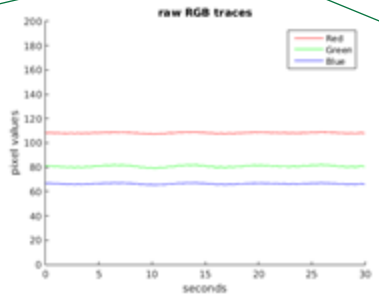
Face detection and tracking

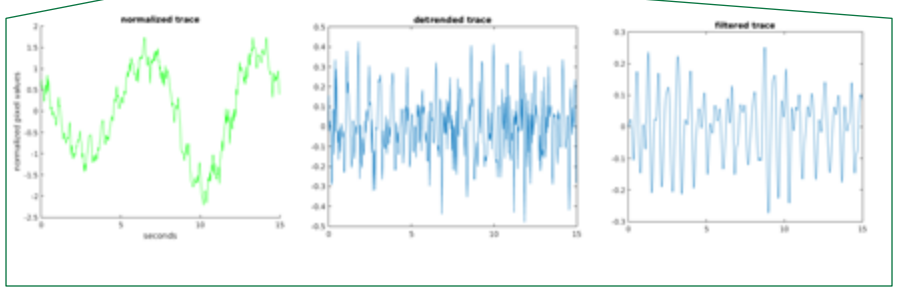
Skin detection





Spatial ROI selection and tracking → Spatial averaging → Color channel selection - combination → Denoising → Physiological parameters estimation





Remote photoplethysmography

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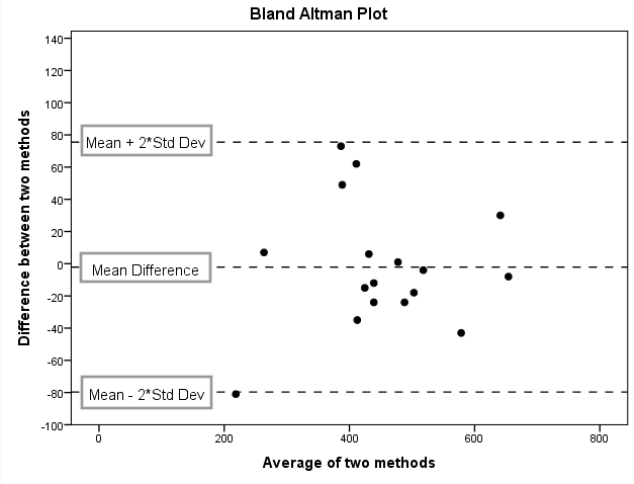
Bland-Altman plot

- Extensively used to evaluate the agreement among two different instruments or two measurement techniques (or one new technique with gold standard)

- If we have *n* measurements pairs
  - S_1 from gold standard
  - S_2 from new measurement technique.
- Each of the n measurements are then represented on the graph by assigning the mean of the two measurements as the abscissa (x-axis) value, and the difference between the two values as the ordinate (y-axis) value.

$$S(x,y) = \left(\frac{S_1 + S_2}{2}, S_1 - S_2\right)$$

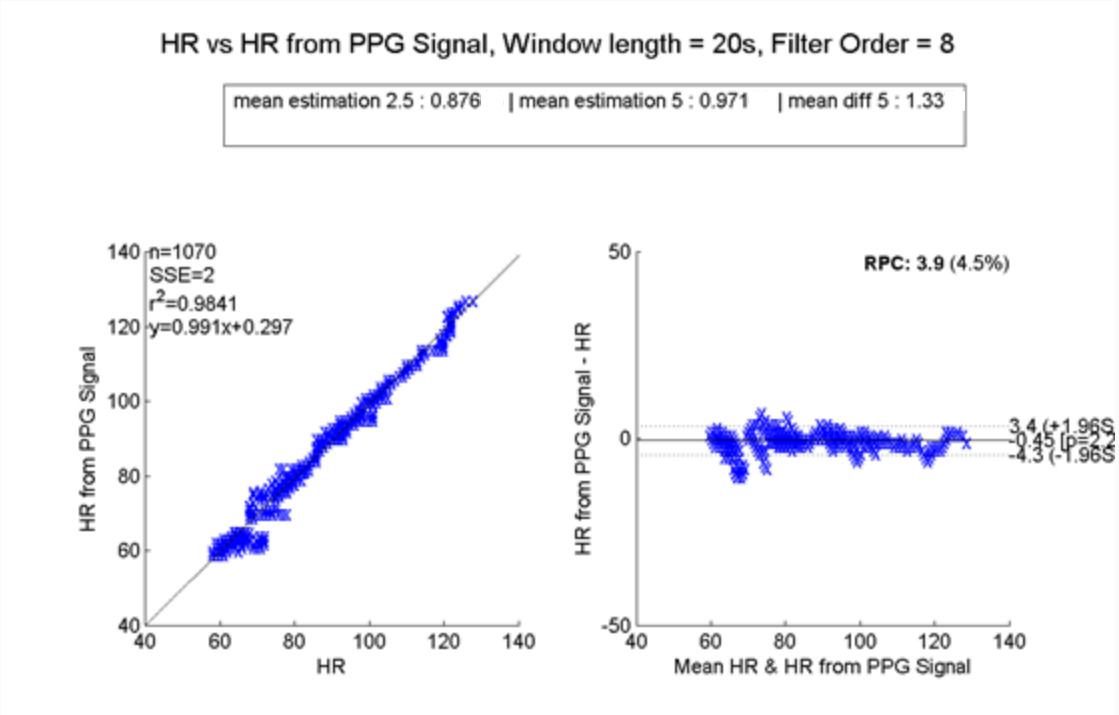


Remote photoplethysmography

Experimental results:

Dataset 1

- 8 simple videos
- Webcam + ambient light
- ~1m



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https://shorturl.at/NCpOs